4 Fig. 4.1 shows a block of mass 0.400 kg attached to a light spring with spring constant of 18.5 N m^{-1} at its equilibrium position. By pulling the block to the lowest point and then releasing it, the block undergoes simple harmonic motion.

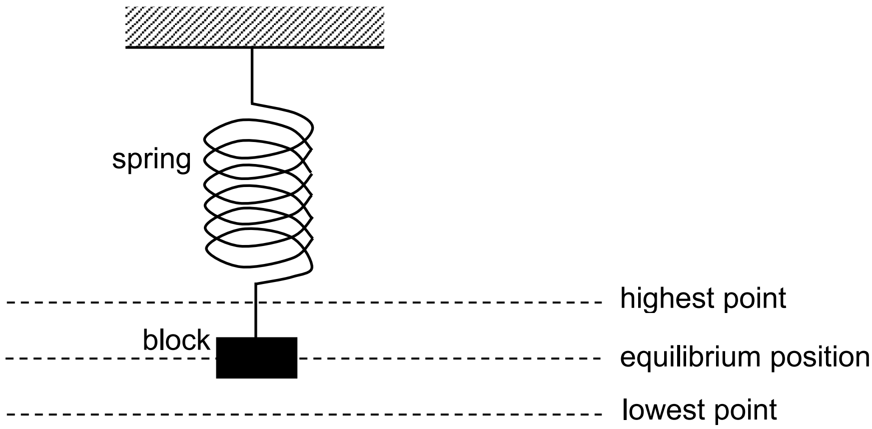

Fig. 4.1

(a) Explain what is meant by *simple harmonic motion*.

....

....

.... [1]

(b) The period T of the oscillations is given by the expression

$$T = 2\pi\sqrt{\frac{m}{k}} \ ,$$

where m is the mass of the block and k is the spring constant of the spring.

(i) Given that the block has a speed of 1.40 m s^{-1} when passing the equilibrium position, determine the amplitude of the oscillations.

amplitude = ... m [2]

(ii) Calculate the maximum tension in the spring during the oscillations.

maximum tension = ... N [3]

(c) (i) On Fig. 4.2, sketch a graph to show the variation with displacement x of the velocity v of the block. Label the graph with appropriate numerical values.

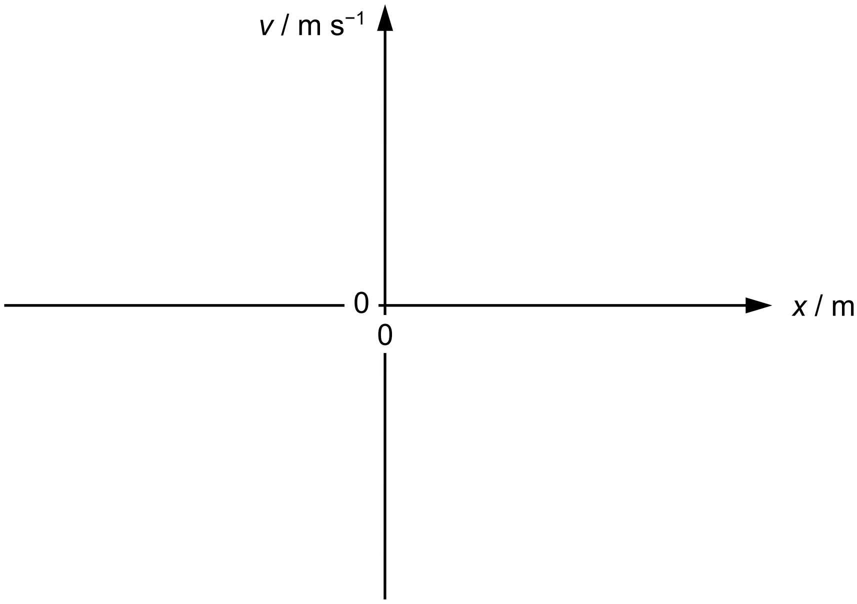

Fig. 4.2

[2]

(ii) The block is now replaced with one of mass larger than 0.400 kg and pulled down by the same amplitude calculated in **(b)(i)**.

On Fig. 4.2, sketch a second graph to show the variation with displacement x of the velocity v of the block with the larger mass.

Label the second graph **Z**. [1]

5 **(a)** A copper wire of diameter 1.8 mm has a current of 17 mA flowing through it.